# Taspol Sawasdee

Computer Engineering and Digital Technology(CEDT)
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## Education

- **Chulalongkorn University** *2023 - Present*
  *B.Eng. in Computer Engineering and Digital Technology (CEDT)* CGPA: 3.67
  - Data Science-Project: Developed an automated system using NLP for main topic extraction from research paper abstracts. The project involved building a robust data pipeline for scraping papers and utilizing clustering to analyze and visualize paper topic categorization, along with data Insight and spatial information.
  - IoT-Project: Developed an IoT system with sensor nodes for environmental monitoring, actuator nodes for automated environment control, and an ML node for swiftlet in–out detection. Designed a system with dynamic gateway election, where a new gateway node is automatically advertised when the primary gateway fails, ensuring continuous connectivity to an MQTT broker and dashboard.

## Working Experience

- **Chulalongkorn University** *September 2025 - December 2025*
  *Teacher Assistance - Embedded System* Thailand
  - Guided students during embedded systems lab sessions, provided technical support for final projects, and oversaw laboratory equipment organization and maintenance.
- **Infinitas by Krungthai** *May 2025 - Agust 2025*
  *Data Scientist* Thailand
  - Review and applied multiarm bandit algorithm techniques to automate marketing campaign selection and improve customer conversion rates.
  - Developed and evaluated python-based algorithms using mock data, performance metrics, and iterative parameter tuning.
  - Built a proof of concept (POC) covering end-to-end campaign workflows, deployed with structured code, docker, and technical documentation.
  - Designed a user interface and presented project outcomes during the company showcase.
- **HiveGround** *May 2024 - July 2024*
  *Roboticist* Thailand
  - Developed and implemented analytical inverse kinematics equations for a 6-DOF robotic arm, validated motion and joint behavior using RViz and Gazebo simulations, and successfully executed the control to achieve accurate end-effector positioning
  - Designed and implemented unit tests for a palm fruit harvesting robotic arm to detect singularity points and analyze the workspace and reachability of the robot arm.
  - Performed a comparison of KDL and IKFast kinematic solvers, visualized the results, and presented findings to determine the optimal approach for a palm fruit harvesting robotic arm.

## Additional Projects

- **CURSR-IV (High Altitude Sounding Rocket)** *June 2025, Texas-USA*
  *Avionics Team - responsible for onboard electronics and flight systems*
  - Participated in the International Rocket Engineering Competition (IREC) as a member of the Chulalongkorn University High Altitude Research Club (CUHAR), contributing to the Avionics subsystem in the design and development of an innovative, functional, and efficient high-altitude (30K feet) sounding rocket.
  - Designed a flight computer PCB with an STM32 microcontroller, addressing power section design, data communication reliability, and system integration, followed by firmware development in STM32CubeIDE
  - Implemented airbrake control logic to control servo actuation using lookup tables generated by the flight dynamics subsystem.
- **FloodNav (Navigation System for Flood Rescue)** *October 2025*
  *Web-Developer*
  - FloodNav is a web application designed to collect and centralize reports during flood incidents, enabling rescue teams to identify casualty locations. The system analyzes collected data from Satellite and Drone imaginary to visualize obstacles, water depth, water current and calculates the shortest and safest routes to support efficient rescue operations.

- Developed a proof of concept using Next.js for the frontend and FastAPI for the backend, supporting three main features: flood reporting, rescue route planning, and resource suggestions, with deployment and demonstration on a self-hosted server.

- **Wild Guild** *September 2025*
  *AI Developer*
  - WildGuide integrates the SEA-LION LLM with curated regional travel datasets from Thailand, Singapore, Vietnam, and Malaysia to generate personalized itineraries, multilingual directions, budget guidance, and real-time safety alerts through a single intuitive interface, while fostering a traveler community to share plans and enrich collective insights.
  - Built a RAG system using Qdrant vector database and bge-m3 embeddings to retrieve relevant trip data from user input, enabling the SEA-LION LLM to analyze context and generate personalized travel plans, then deployed the system API on GCP.

- **OFFIX (Office Syndrome Prevention Application)** *August 2024*
  *Research and Development Team*
  - OFFIX aims to prevent office syndrome among frequent computer users by using webcam-based pose estimation to monitor sitting posture, notify users of ergonomically incorrect positions, and encourage healthy habits through exercise guidance and a reward system.
  - Developed a real-time posture monitoring pipeline using Pose Estimation Model(BlazePose) for keypoint extraction and rule-based classification, integrated into the OFFIX application.
  - Developed a posture monitoring model for lateral-view posture prediction, improving classification performance through ergonomics-based rules.

## Technical Skills and Interests

**Programing Languages (Intermediate - Advance)**: Python, SQL, C/C++, Java, Typescript, JavaScript
**Developer Tools**: VS Code, Git/GitHub, Docker, Next.js, Express.js, Generative AI, MiniConda, Package Manager(uv,pnpm)
**AI Tools**: PyTorch, TensorFlow, scikit-learn, OpenCV, MediaPipe, Hugging Face
**Data Tools**: NumpPy, Pandas, PySpark, Plotly, Matplotlib, Streamlit, Jupyter Notebooks, Google Sheet
**Cloud/Databases**: Amazon Cloud, Google Cloud, Alibaba Cloud, MongoDB, PostgreSQL, Qdrant
**Embedded System**: STM32CubeIDE, Platformio, Kicad, ArduinoIDE, MQTT Broker, STM32, ESP32, Jetson nano
**Soft Skills**: Eager to learn new things, Persistent and resilient, Responsible and reliable
**Coursework**: Computer Programming, Computer Engineering Mathematics, Intro to Data Science, Embedded System, Software Engineering, AI/ML, Networks
**Areas of Interest**: Data Science, Data Engineer, Networks, IoT System, DevOps, Robotics

## Languages

- **Thai,** Native
- **English,** Working Proficiency
- **Chinese,** HSK3

## Positions of Responsibility

- **Head of Operation Department,** CEDT Sandbox club *Sep 2025 - Jan 2026*
- **Avionics Subsystem Member,** CU High Altitude Research club *Aug 2024 - July 2025*
- **Lead of Venue Department,** CEDT Innovation Summit 2025 *Dec 2024 - July 2025*
- **Operation Department Member,** CEDT Sandbox club *Sep 2024 - Sep 2025*
- **CU Photo club Member,** CU Photo club *Aug 2024 - Present*
- **Research and Development Team,** OFFIX Solution *2023 - Present*

## Achievements

- **Honorable Mention**: Ford Smart Mobility Challenge 2025 Grant 40,000 Baht *2025*
- **Finalist**: Pan-SEA AI Developer Challenge 2025 *2025*
- **3rd Place**: 30K COTS category in International Rocket Competition *2025*
- **100,000 Funding**: Awarded by TED Youth Startup for OFFIX AI Posture Detection *2024*
- **1st Place**: The GISTDA School Satellite Competition *2024*
- **1st Place**: Digital Youth Network Thailand Grant 100,000 Baht *2023*